

Requirement Analysis Phase Template

DATA FLOW DIAGRAMS

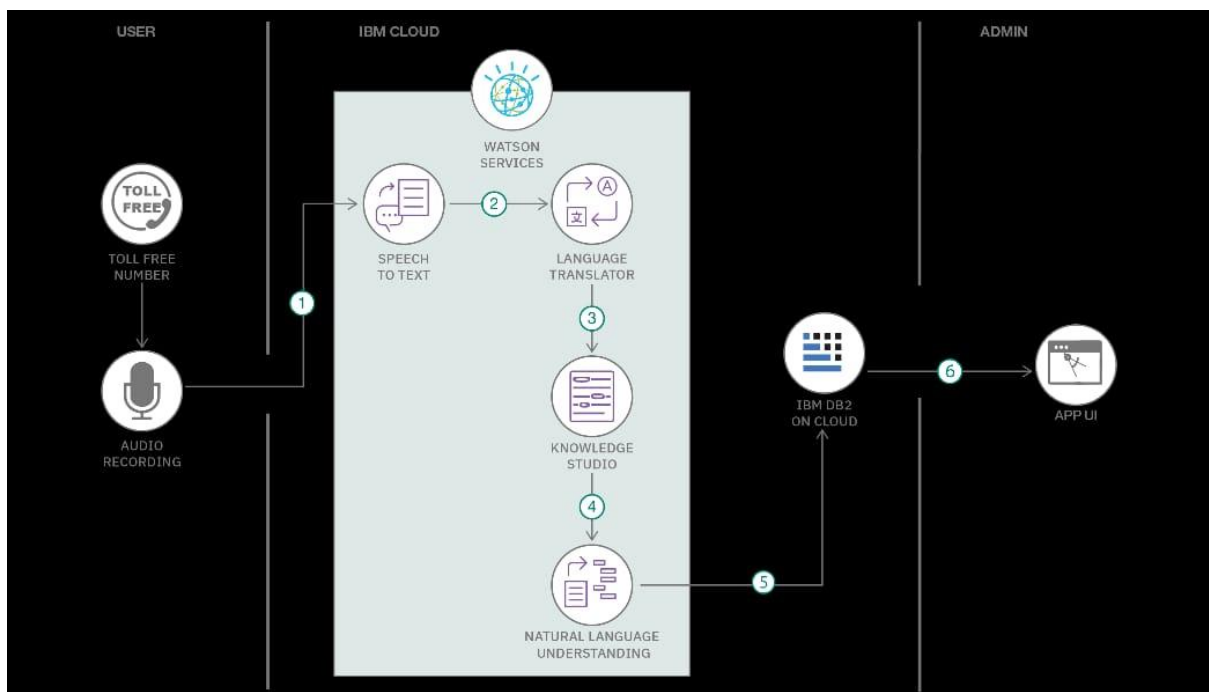
DATE:	27/09/2026
TEAM ID:	SWTID-2026-6715
PROJECT NAME:	AI FAQ Assistant API
MAXIMUM MARKS:	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table 1: Components & Technologies.

S.No	Component	Description	Technology
1	User Interface	User registration, login, FAQ search and AI Library answers.	HTML, JavaScript
2	Application Logic-1	User registration, login and authentication.	Node.js, Express.js, JWT, bcryptjs
3	Application Logic-2	Create, view, update and delete FAQs.	Node.js, Express.js
4	Application Logic-3	Generate AI answers and topic suggestions.	Google Gemini API
5	Database	Store user details and FAQ records.	MongoDB, Mongoose
6	Cloud Database	Store application data online, if used.	MongoDB Atlas

Table 2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Uses open-source frameworks for frontend, backend and database operations.	Express.js, Node.js, MongoDB
2	Security Implementations	Uses password hashing, JWT authentication and protected routes.	bcryptjs, JWT, HTTP
3	Scalable Architecture	Modular REST API architecture supports adding more users and FAQs.	Node.js, Express.js, MongoDB
4	Availability	Provides access to published FAQs and AI-assisted answers through the application.	Node.js Server / Cloud Hosting
5	Performance	Uses database queries and semantic search to retrieve relevant FAQs and generate answers.	MongoDB, Mongoose, Google Gemini API